

MOHAMED HAMDOUN

AI SYSTEMS DEVELOPER / APPLIED AI RESEARCHER / CREATIVE TECHNOLOGIST

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GitHub github.com/mohamed2hamdoun **Portfolio** mohamed2hamdoun.github.io

PROFILE

Computer Science undergraduate specializing in Artificial Intelligence and Machine Learning, with hands-on experience in backend development, agentic and LLM-powered systems, and applied ML research. I design intelligent systems and digital experiences that turn complex ideas into useful, memorable products — from auditable multi-agent architectures to a peer-reviewed ML study. A background in filmmaking and visual storytelling shapes how I make technical systems communicate clearly.

EDUCATION

Bachelor of Computer Science in AI & Machine Learning (BCSAI) Expected 2028
University of Fujairah — United Arab Emirates

EXPERIENCE

Chatbot Developer Intern — Flora Group W.L.L 2 months
Built and enhanced an AI-powered chatbot supporting logistics and HR systems, integrated with real-world internal and external APIs.

- Implemented LLM-based conversational logic using LangChain and LangGraph.
- Integrated chatbot functionality with internal and external APIs to handle manager-level requests.
- Built Python backend components and applied observability with Langfuse.
- Worked across authentication, role-based requests, and client-interaction workflows.

Stack: Python · LangChain · LangGraph · FastAPI · Langfuse · REST APIs

FLAGSHIP PROJECTS

Agentic AI for Housing Arrears Hackathon — Top 3, Awarded
A multi-agent system turning a multi-day housing-arrears rescheduling process into an immediate, auditable AI-assisted service — intake validation, document intelligence/OCR, financial and hardship analysis, deterministic policy logic, and human escalation.
Stack: Python · LangGraph · FastAPI · Pydantic · OCR · Ollama · github.com/mohamed2hamdoun/sahel_agent

Hermes — AI Design Intelligence Platform In development
A local-model platform that researches visual references, extracts design intelligence, generates websites, critiques its own output, and improves through iterative feedback — an end-to-end design loop running on local Qwen.
Stack: Python · Qwen (local) · Ollama · FastAPI · TypeScript · Embeddings

AI Book Summarization & Knowledge System Working system
Processes books page by page into story-driven, visually structured Obsidian knowledge vaults — page-level citations, Mermaid diagrams and mind maps, with resume recovery and failure isolation for long runs.
Stack: Python · Ollama · OCR · Mermaid · Obsidian

ChatGPT-like Web Application Solo
A conversational web backend on FastAPI and LLM APIs — prompt handling, conversation flow, chat-history management, and scalable real-time endpoints.
Stack: Python · FastAPI · LLM APIs

RESEARCH

Probabilities Feature Enrichment for Improved Early Diabetes Prediction Published — Algorithms
Peer-reviewed study enriching clinical feature spaces with class-conditional probability features from Gaussian Mixture Models and Kernel Density Estimation, combined with PCA and correlation reduction, to improve early diabetes prediction.
Methods: Logistic Regression · GMM · KDE · PCA · Feature enrichment · Model evaluation

APPLIED ML & OPEN SOURCE

California House Price Prediction End-to-end regression with an XGBoost regressor — analysis to evaluation. /houses-prices-prediction	Rocks vs Mines Prediction Full classification pipeline — preprocessing, EDA, feature engineering. /ml-rocks-mines-prediction
Loan Status Prediction Supervised classification of loan approval with model comparison. /Loan-Status-Prediction...	Fake News Prediction NLP text-classification with preprocessing + feature extraction. /fake-news-prediction
AI-Powered Applications (IBM) Applied AI projects from IBM's "Building AI-Powered Applications" course. /Building-AI-powered-applications-IBM-course	

CERTIFICATIONS

IBM AI Engineering — Coursera	2025	IBM Generative AI Engineering — Coursera	2025
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Front-End Development — Meta / Coursera	2025	CS50x: Programming with Python — Harvard	2023
Google IT Automation with Python — Coursera	2022		

TECHNICAL SKILLS

AI & BACKEND	Python · FastAPI · Pydantic · SQLAlchemy · PostgreSQL · REST APIs · LangChain · LangGraph · LLM APIs · Langfuse · Ollama · Qwen
MACHINE LEARNING	Scikit-learn · XGBoost · PyTorch · Pandas · NumPy · Matplotlib · Seaborn · Model evaluation · Experimental design
WEB	TypeScript · React · Next.js · Tailwind CSS · Three.js · HTML · CSS · JavaScript
CREATIVE	Premiere Pro · After Effects · Photoshop · Filmmaking · Motion graphics · Visual storytelling